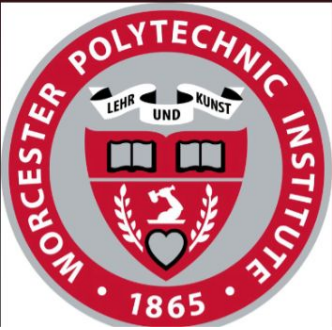




Statistical Artificial Intelligence LAB
SAIL Lab

SAIL

Diffusing the Unknown

Bayesian Inverse Problem with Diffusion Models

Alexander Kaneko

SAIL Lab

Worcester Polytechnic Institute

Bayesian Inverse Problems

Inverse problems

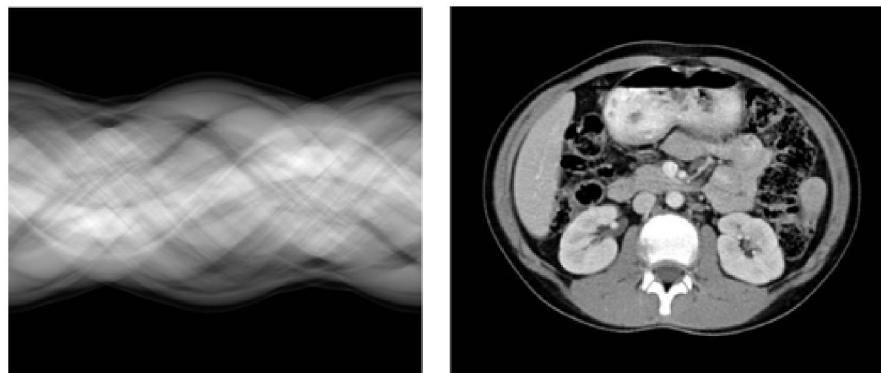
- If we observe a noisy image/signal, can we reconstruct a clean one?

Noisy $\longrightarrow$ Clean



Deblurring images

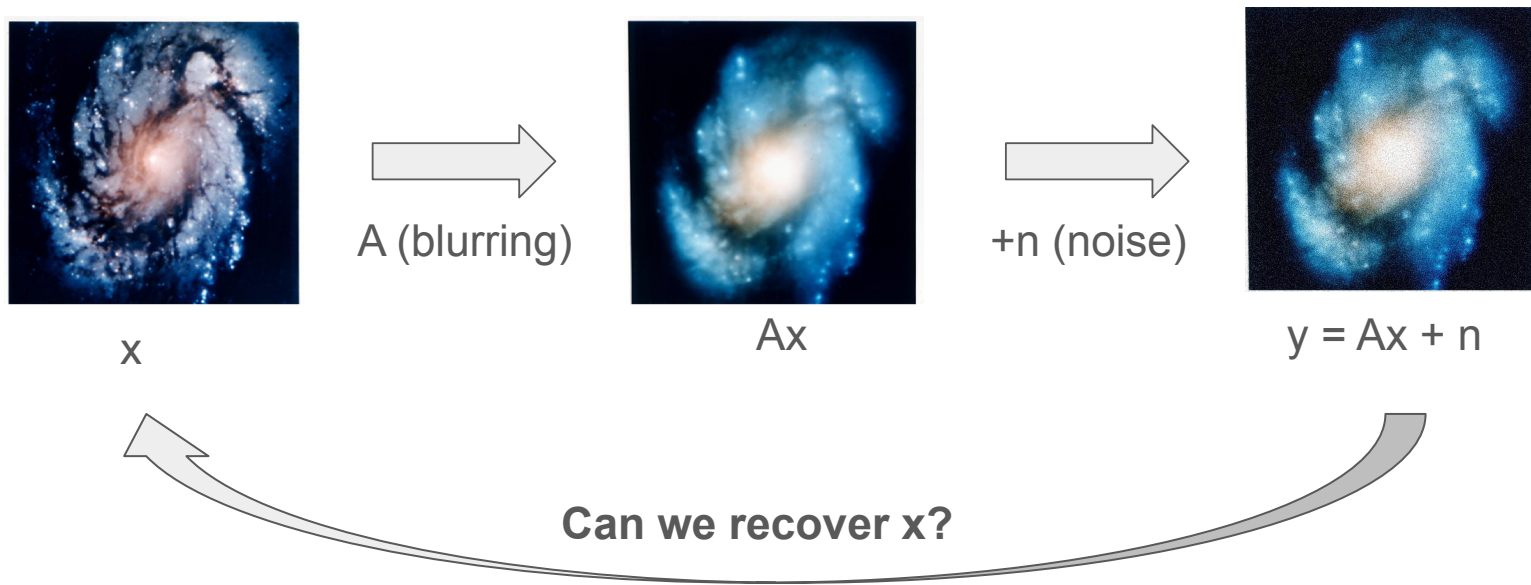
Noisy $\longrightarrow$ Clean



CT scan of organ

Mathematical form

- $y = Ax + n$

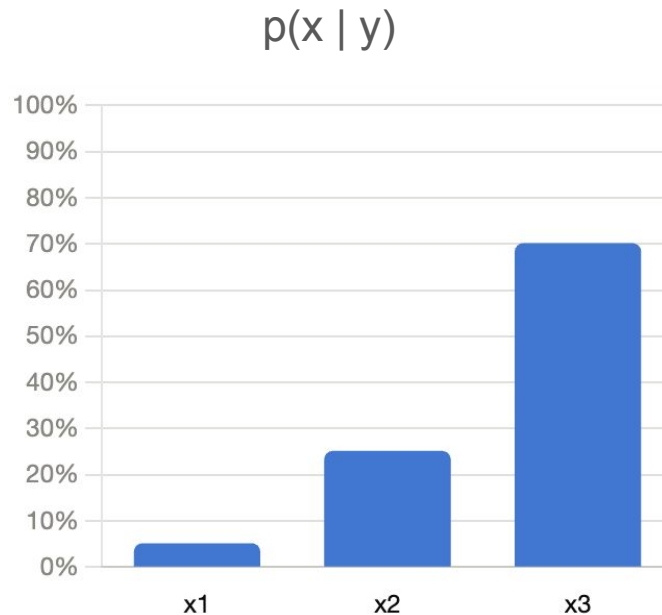
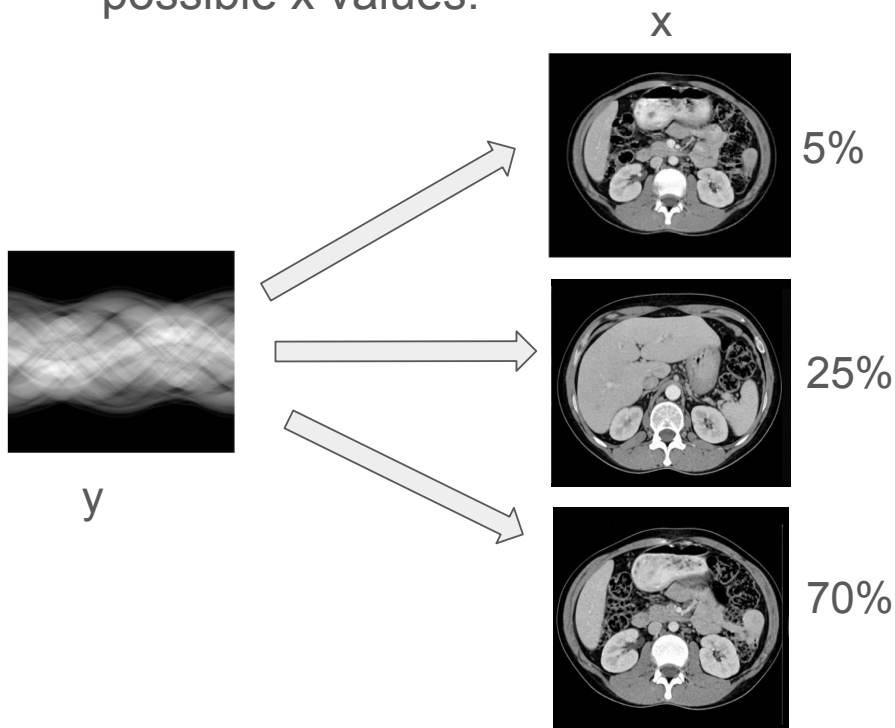


Why finding x is difficult

- There may be no solutions, multiple solutions, or unstable solutions
- Computing A^{-1} directly could be possible if A is a linear operator
- Doing so is costly and not possible if A is non linear.

Bayesian inverse problems

- Rather than recovering one singular value, we find the **distribution** of the possible x values.

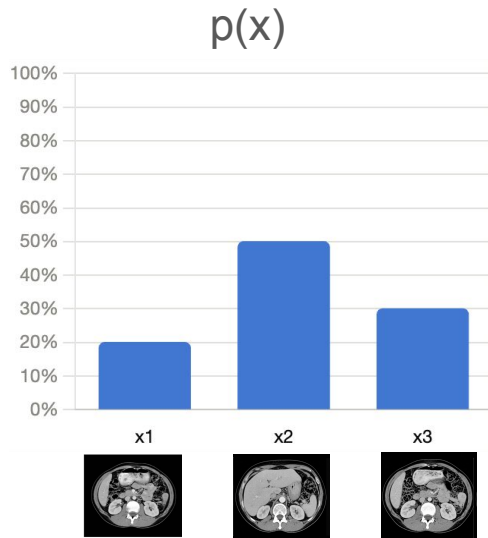


Finding $p(x | y)$

- From Bayes theorem, $p(x | y) = p(x) * p(y | x) / p(y)$
- $p(y)$ can be treated as a constant
- Thus, computing $p(x) * p(y | x)$ and normalizing gives us $p(x | y)$

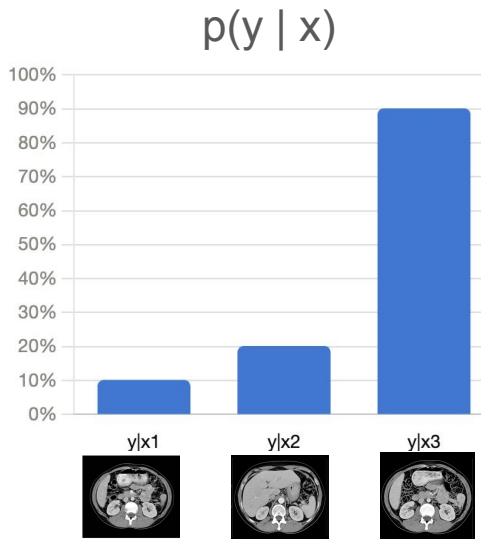
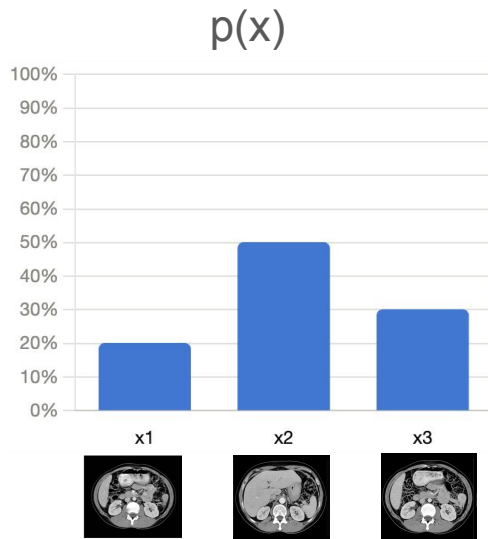
Finding $p(x | y)$

- $p(x)$ is the **prior**, the distribution of how often the images appear naturally



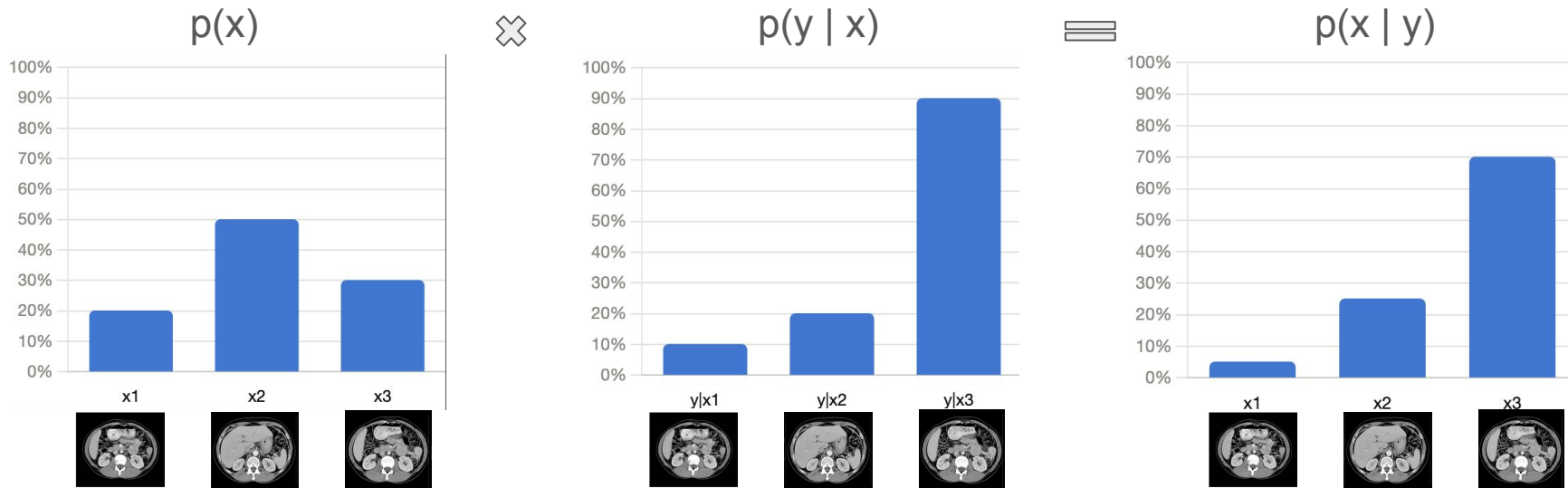
Finding $p(x | y)$

- $p(y | x)$ is the **likelihood**, how likely y is equal to $Ax + n$ given an x



Finding $p(x | y)$

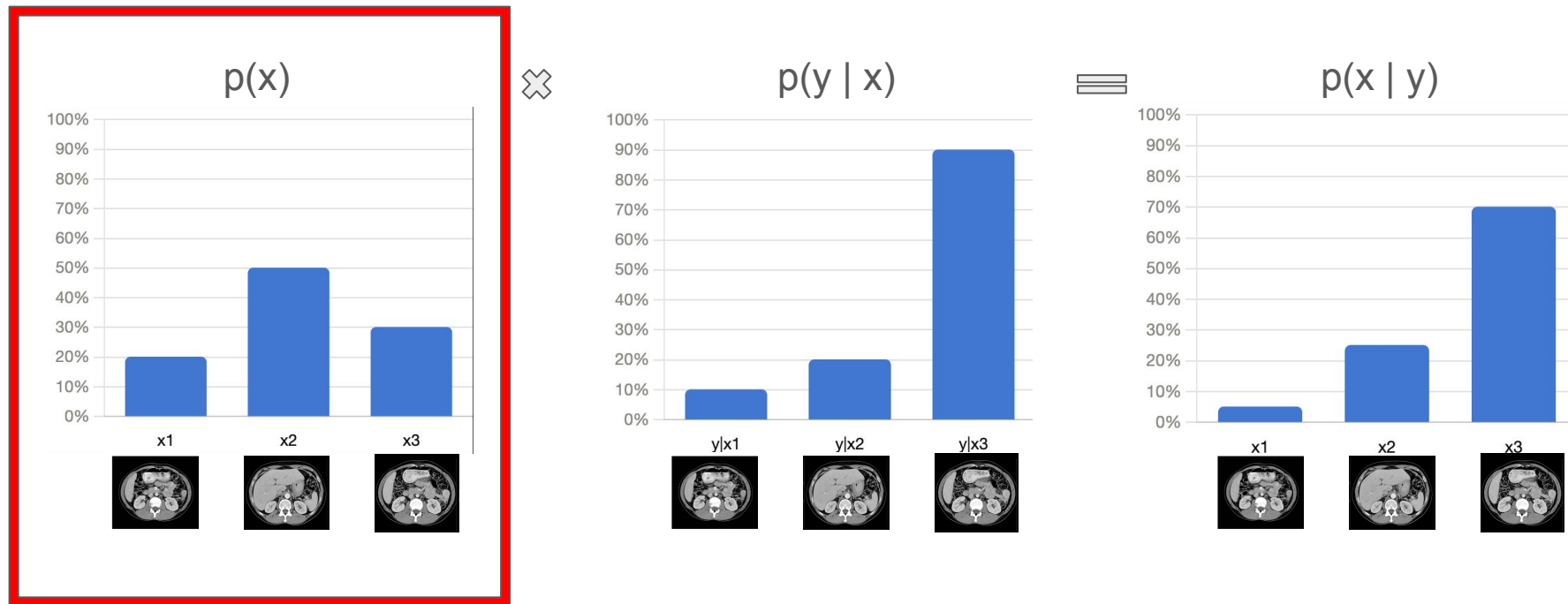
- $p(x | y)$ is the **posterior**, found by multiplying the prior by the likelihood and normalizing



Finding $p(x | y)$

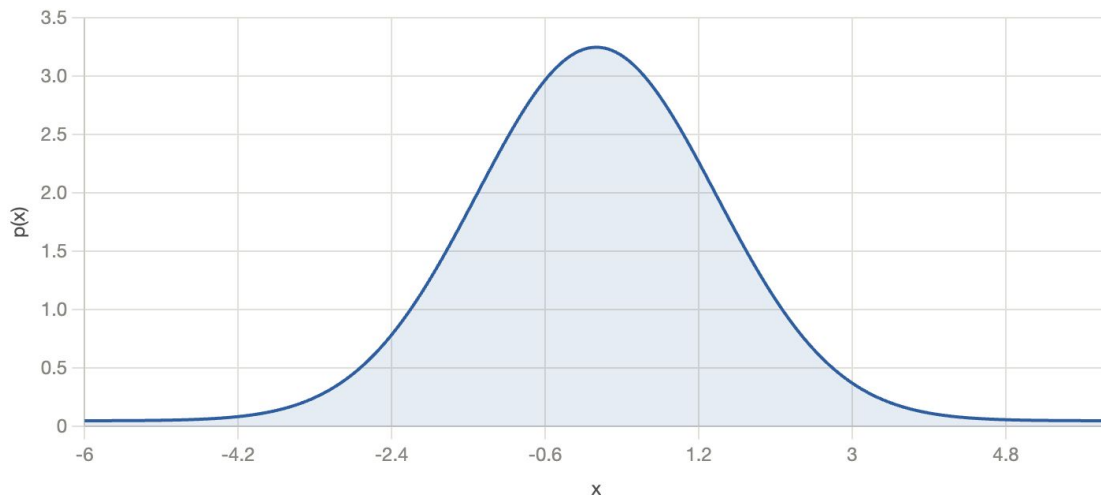
$$y = \mathcal{A}(x) + \eta, \quad \eta \sim \mathcal{N}(0, \sigma_y^2 I),$$
$$p(y | x) \propto \exp\left(-\frac{\|y - \mathcal{A}(x)\|_2^2}{2\sigma_y^2}\right).$$

- $p(y | x)$ is relatively easy to compute, $p(x)$ is the difficult part



Modeling the prior $p(x)$

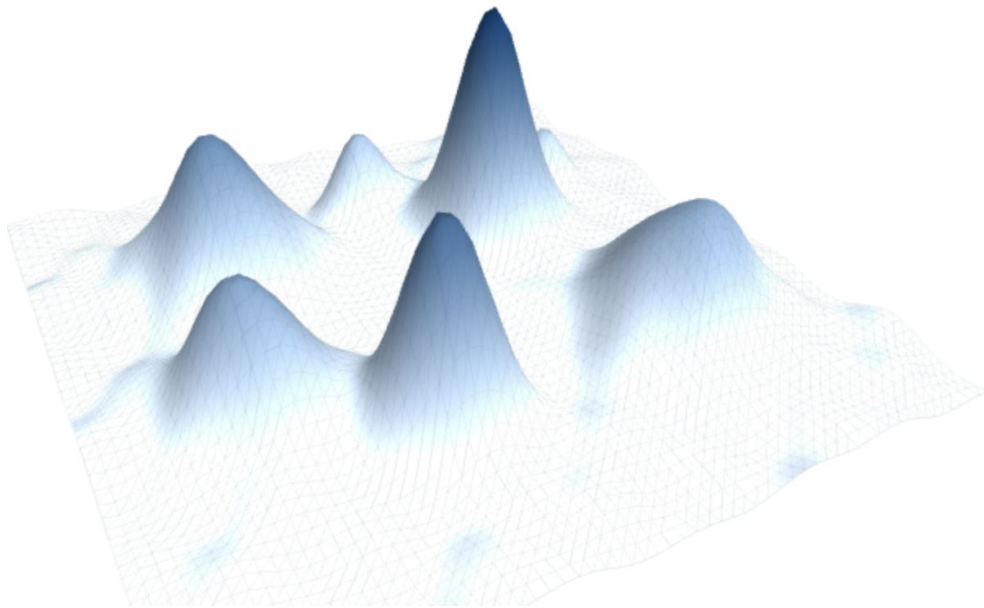
- One common way we can model $p(x)$ is by setting it to a **fixed** distribution, such as a Gaussian distribution.



Examples: blood temperature, bone density, heart rate at rest

Modeling the prior $p(x)$

- But for more complex distributions, $p(x)$ needs to be **learned**.

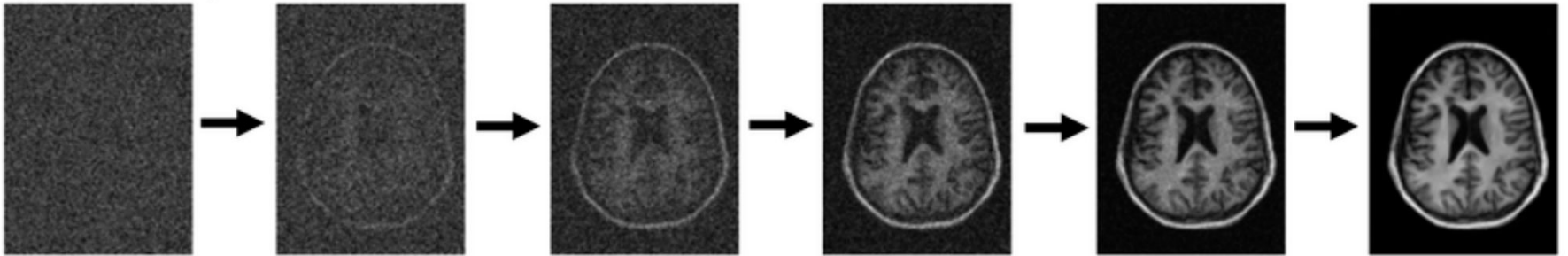


Examples: natural images, CT scans, protein structures

Diffusion models

Diffusion models

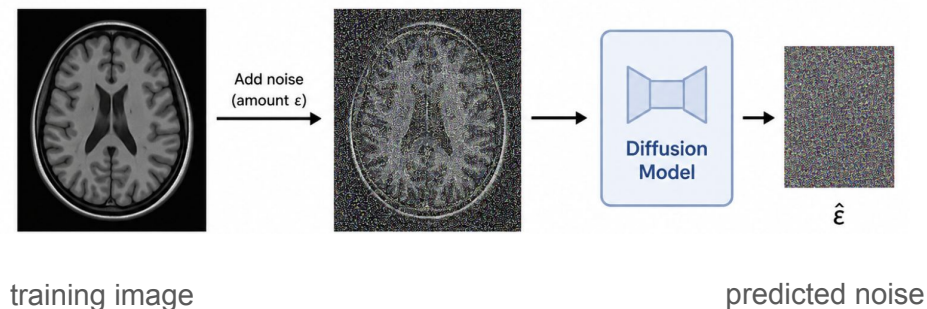
- Known for high-quality image generation (DALL-E 2, Midjourney)
- They create clean images by iteratively removing noise (more details in Appendix)



Generating an image

Diffusion models

- The model is trained by adding noise ϵ to given image, and then asking a neural network to predict ϵ .
- Learning to predict ϵ corresponds to learning the value of $\nabla_x \log p_t(x)$, known as the **score**.



$$s_{\theta}(x_t, t) \approx \nabla_{x_t} \log p_t(x_t) = -\frac{\epsilon_{\theta}(x_t, t)}{\sqrt{1 - \bar{\alpha}_t}}.$$

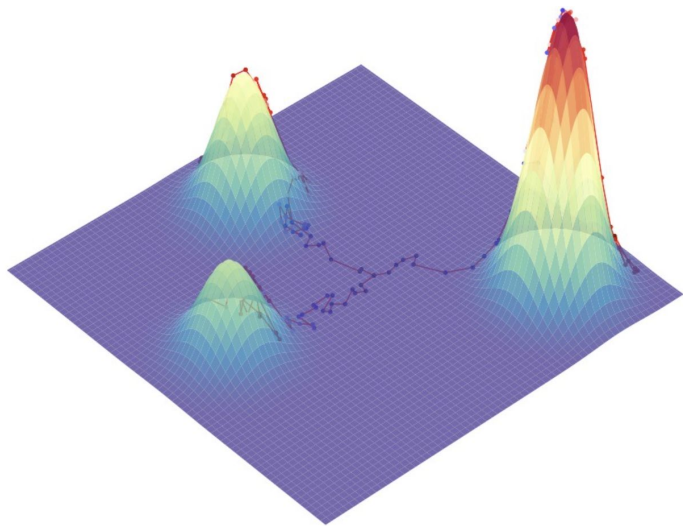
neural
network
prediction

score

noise

Diffusion models

- Through a process called **Langevin dynamics**, we are able to use the score to find our desired $p(\mathbf{x})$.



$$\mathbf{x}_{t+1} = \mathbf{x}_t + \tau \nabla_{\mathbf{x}} \log p(\mathbf{x}_t) + \sqrt{2\tau} \epsilon_t$$

Diffusion models for inverse problems

H. Chung et al., “Diffusion Posterior Sampling for General Noisy Inverse Problems,” ICLR, 2023.

A. Güngör et al., “Bayesian Conditioned Diffusion Models for Inverse Problems,” arXiv, 2024.

Posterior-score decomposition

- In practice, we try to learn the score of $p(x | y)$ instead of finding it directly.
- Using Bayes Theorem we can decompose it into the sum of a **diffusion prior** and measurement guidance terms.

$$\nabla_{x_t} \log p_t(x_t | y) = \underbrace{\nabla_{x_t} \log p_t(x_t)}_{\text{diffusion prior}} + \underbrace{\nabla_{x_t} \log p_t(y | x_t)}_{\text{measurement guidance}} .$$

This is what a diffusion
models learns!

How DPS approximates the likelihood score

- The **measurement guidance** term (score of likelihood) is generally intractable, so an approximation is needed.
- DPS (Chung et al., ICLR 2023) utilizes a clever trick with Tweedies identity to approximate it tractably (detailed derivation in the Appendix).

$$\nabla_{x_t} \log p_t(x_t \mid y) = \underbrace{\nabla_{x_t} \log p_t(x_t)}_{\text{diffusion prior}} + \underbrace{\nabla_{x_t} \log p_t(y \mid x_t)}_{\text{measurement guidance}} .$$

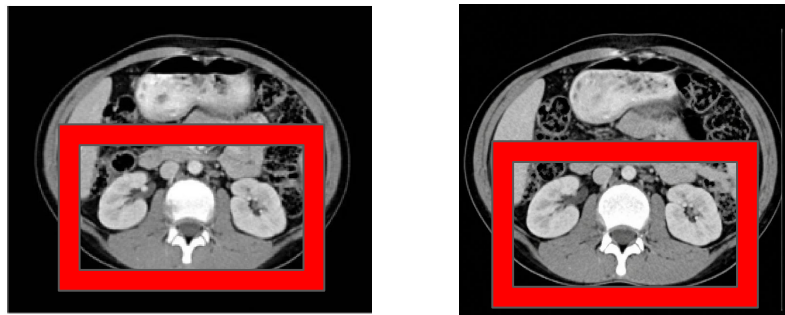
Results for Diffusion Posterior Sampling

- Using this method, we are able to “sample” from the posterior distribution $p(x | y)$ as shown below.
- DPS consistently produces high-quality reconstructions that closely match the ground truth, avoiding noise-amplification artifacts commonly found.



What makes the solution Bayesian?

- We can sample multiple times to get multiple possible solutions, constructing our the distribution of possible x values.
- We can find the posterior mean, perform uncertainty quantification, or find visibly different reconstructions both compatible with y .

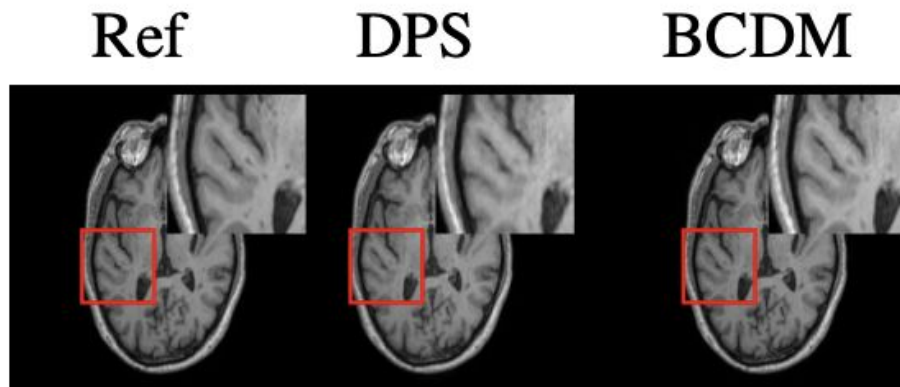


Shared region: high confidence

Bayesian Conditioned Diffusion Models

- Instead of heuristically splitting up the score of the posterior, (Güngör et al., 2024) learn it directly through a diffusion model. This requires separate training for each specific inverse problem

$$s_{\phi}(x_t, y, t) \approx \nabla_{x_t} \log p_t(x_t \mid y)$$



BCDM achieved improvements over DPS based off several metrics (pSNR, SSIM, FID)

Tradeoffs and Limitations

Question	DPS	BCDM
Learned quantity	Unconditional score $s_\theta(x_t, t)$	Conditional score $s_\phi(x_t, y, t)$
Measurement use	Incorporated during inference	Incorporated during training and inference
Training data	Clean samples x	Clean samples and simulated measurements from the selected operator
New measurement y	No retraining	No retraining
New operator $\mathcal{A}$	Often usable without retraining	Usually requires a new network
Operator class in the paper	Differentiable linear and nonlinear	Linear
Noise models in the paper	Gaussian and Poisson	Gaussian
Main approximation	Noisy likelihood score through $\hat{x}_0(x_t)$	Neural approximation of the conditional score and diffusion discretization under the stated model
Main computational cost	Repeated score evaluations and likelihood backpropagation	Task-specific training and repeated evaluation of an unrolled conditional network

Table comparing the two methods

Takeaways

- Bayesian inverse problems seek a distribution of plausible solutions,

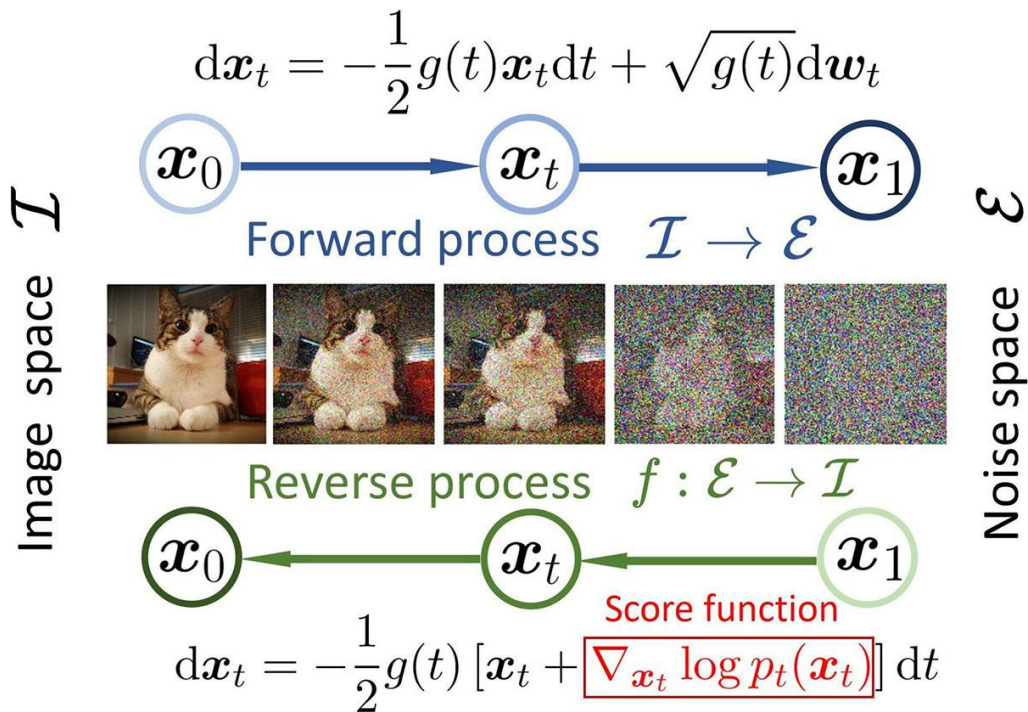
$p(x | y) \propto p(y | x)p(x)$, not only a single reconstruction.

- Diffusion models provide a learned, time-dependent prior score for complex data distributions.
- Posterior samples should be measurement-consistent and accompanied by an assessment of uncertainty.

Future directions slide

- How close are samples from DPS and BCDM to the true posterior distribution?
- How robust are the methods when the forward operator or measurement-noise distribution is misspecified?
- Can posterior sampling require fewer neural-network evaluations without losing reconstruction quality or sample diversity?

Appendix 1



More detailed depiction of diffusion models

Appendix 2

DPS first estimates the clean variable through Tweedie's identity.

$$\hat{x}_0(x_t) = \frac{x_t + (1 - \bar{\alpha}_t)s_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}}.$$

It then uses the approximation

$$\nabla_{x_t} \log p_t(y \mid x_t) \approx \nabla_{x_t} \log p(y \mid \hat{x}_0(x_t)).$$

For Gaussian observations

$$\nabla_{x_t} \log p(y \mid \hat{x}_0(x_t)) = -\frac{1}{2\sigma_y^2} \nabla_{x_t} \|y - \mathcal{A}(\hat{x}_0(x_t))\|_2^2.$$

$$x_t \longrightarrow \hat{x}_0(x_t) \longrightarrow \mathcal{A}(\hat{x}_0(x_t)) \longrightarrow \|y - \mathcal{A}(\hat{x}_0(x_t))\|_2^2.$$

Flow of approximation